

Guanzhong Wu, Ph.D.

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SUMMARY

Ph.D. physicist and product development engineer specializing in MRAM and emerging non-volatile memory. Expertise spanning MTJ stack design, nanoscale device physics, memory test development, and reliability qualification. Proven record of cross-functional collaboration with foundry and circuit-design teams to improve memory performance, yield, and reliability, backed by 9 peer-reviewed publications in magnetics and spintronics.

TECHNICAL SKILLS

Memory & Devices: MRAM, MTJ stack design & characterization, nanoscale device-physics simulation, BER analysis, yield monitoring

Test & Qualification: Functional testing; reliability qualification (endurance, retention, HTOL, HAST)

Hardware & Tools: FPGA-based memory test vehicle design

Measurement & Characterization: Ferromagnetic resonance, scanned-probe microscopy, spin-torque oscillators

PROFESSIONAL EXPERIENCE

Manager, MTJ Device and Product Characterization and Testing Aug 2025 – Present

Product Development Engineer | *Avalanche Technology Inc., Fremont, CA* Jun 2022 – Aug 2025

- Collaborate across foundry and circuit-design teams to optimize MRAM performance by designing tests, analyzing results, monitoring yield, and characterizing bit error rate (BER).
- Execute MRAM functional testing on package die and wafer.
- Lead reliability qualification campaigns including endurance, retention, HTOL, and HAST.
- Design and characterize MTJ stacks and run nanoscale device-physics simulations to guide stack development.
- Develop FPGA-based memory test vehicles to validate new device designs.

Postdoctoral Associate | *New York University, New York, NY* Jan 2022 – Jun 2022

- Developed spin-torque-driven antiferromagnetic oscillators for next-generation high-frequency spintronic devices.

Graduate Research Assistant | *The Ohio State University, Columbus, OH* Dec 2017 – Dec 2021

- Investigated nanoscale magnetization dynamics and magnetic anisotropy using a high-sensitivity scanned-probe ferromagnetic resonance force microscope.
- Authored and co-authored multiple peer-reviewed publications on magnetic interfaces and spin dynamics.

Graduate Teaching Assistant | *The Ohio State University, Columbus, OH* Aug 2016 – Nov 2017

- Instructed undergraduate physics laboratory and recitation sections, supporting student learning and assessment.

Research Assistant | *Peking University, Beijing, China* Aug 2014 – Jul 2016

- Investigated chemical vapor deposition of vertically standing graphite for nanomaterial applications.

EDUCATION

Ph.D., Physics — The Ohio State University 2021

Dissertation: Scanning Ferromagnetic Resonance Force Microscope Study of the Interface Between $Y_3Fe_5O_{12}$ and Nonmagnetic Materials. Advisor: P. Chris Hammel

M.S., Physics — The Ohio State University 2018

B.S., Physics — Beihang University 2014

SELECTED PUBLICATIONS

9 peer-reviewed publications in Nature Communications, Nano Letters, Physical Review B, and IEDM (selected below):

- Z. Wang, Z. Wei, G. Wu, et al. International Electron Devices Meeting (IEDM), IEEE, 2023.
- H. Ren, X. Y. Zheng, S. Channa, G. Wu, et al. Nature Communications 14.1 (2023): 1406.
- G. Wu, et al. Nano Letters 22.3 (2022): 1115–1121.
- G. Wu, et al. Applied Physics Letters 118.4 (2021): 042403.
- Y. Cheng, A. J. Lee, G. Wu, et al. Nano Letters 20.10 (2020): 7257–7262.
- G. Wu, et al. Physical Review B 101.18 (2020): 184409.
- J. S. Jamison, Z. Yang, B. L. Giles, J. T. Brangham, G. Wu, et al. Physical Review B 100.13 (2019): 134402.
- J. Xu, S. Singh, J. Katoch, G. Wu, et al. Nature Communications 9.1 (2018): 1–6.
- Y. K. Luo, J. Xu, T. Zhu, G. Wu, et al. Nano Letters 17.6 (2017): 3877–3883.